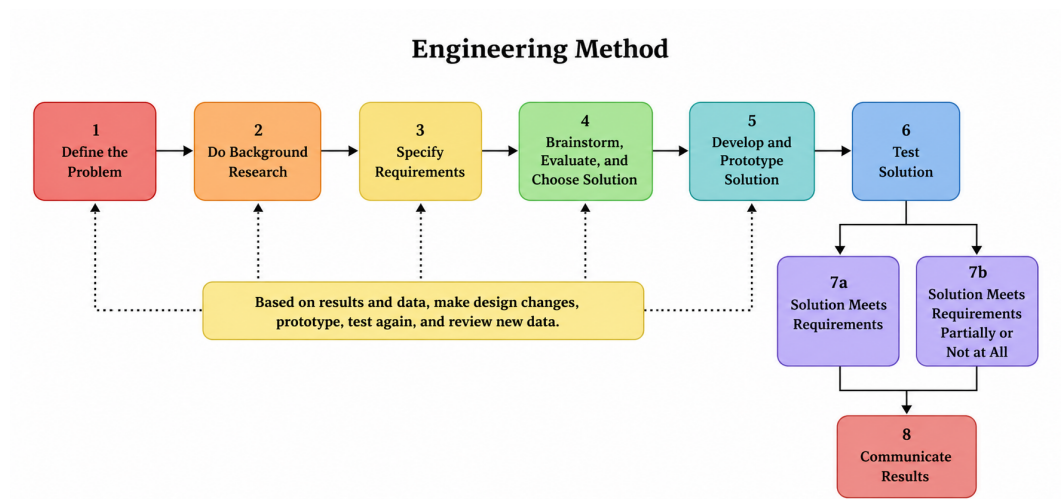


COMP6733 IoT Research Project

26T2 Version 1.0

1 Overview

The project component of COMP6733 aims to give you an opportunity to solve a real-life problem in IoT. The format of the project aims to emulate the classical engineering design process, which is depicted in the below figure. This project will therefore test your ability to solve problems, propose solutions and analyse their suitability as well as your skills in implementing a solution, which includes planning, testing and programming. The scope of the project is open to allow you to explore the subject area.



Section 2 of this document describes a number of projects that you may choose to work on; however, you are free to propose your own project. You are expected to do the project in a team of five students and bid for these projects by expressing your preferences. The Lecturer-in-Charge (LiC) will then allocate the projects according to your expressed preferences by Week 3. More details on the bidding process are described in Section 3.2.

The project consists of two stages. In the first stage, you should consider different options that you can use to solve the problem. You do this by analysing the problem and researching possible solutions. Based on your research and analysis, you are required to propose how you plan to tackle the problem. You are required to submit a preliminary report and give a project presentation (in Week 5) to explain your proposed solution. Your report should include:

1. A discussion on how you plan to tackle the problem with your justification.
2. The implementation that you plan to do with a timeline.

3. The goals of your demonstration. You will need to specify goals for what you will be demonstrating in the final demonstration.

The second stage of the project will be devoted to working towards the goal that you have set for yourselves.

The LiC will go through your preliminary report and provide feedback on the work that you plan to do. In all cases, you should be aware the University expects students taking a 6 UoC course to devote about 14 hours per week to each course. Given the above expectations on the amount of time you will be spending on the project and to ensure that all projects meet a common standard, the LiC may choose to add, remove or change tasks from your proposed plan.

2 Project Description

The proposed projects are organised into five thematic categories. Each project includes a short description, a hardware/scope outline, and reference papers or links for background reading. You are encouraged to follow the cited references and explore further on your own. **If hardware listed for a project is not available in the lab, please discuss with the LiC as early as possible.**

2.1 Category A: Agritech & Environment

2.1.1 Project A1: LLM-Guided Multispectral Crop Health Monitor

Vertical farms typically use pre-determined schedules for nutrient and light delivery to plants, struggling to dynamically alter resource inputs. Industrial multispectral cameras cost >\$1k, while the AS7265x is a cost-effective sensor providing 18 discrete VIS/NIR channels covering the photosynthesis light spectrum. In this project, you will design an Arduino-based system to monitor leaf reflectance spectra, then use a cloud-based LLM (e.g. Claude/GPT API) to determine plant stress in real time, and push actionable results over MQTT to AWS IoT.

References:

- Špundová, Martina, et al. "What to choose for estimating leaf water status—spectral reflectance or in vivo chlorophyll fluorescence?." *Plant Phenomics* 6 (2024): 0243.

2.1.2 Project A2: ThermoLeaf—Closed-Loop Irrigation with Thermal Imaging

Leaf-to-air temperature difference (ΔT) is a proven proxy for plant water stress: as transpiration cools the leaf, water-stressed leaves run hotter than ambient. Commercial IR cameras are

costly, but the FLIR Lepton makes thermal imaging affordable. You will build a Raspberry-Pi-based field node with a FLIR Lepton + ambient temperature sensor that displays a ΔT map of a plant, triggers irrigation when ΔT exceeds a threshold, and evaluates correlation (R^2) between ΔT and a soil-moisture sensor as ground truth.

References:

- Costa, J. Miguel, Olga M. Grant, and M. Manuela Chaves. "Thermography to explore plant–environment interactions." *Journal of experimental botany* 64.13 (2013): 3937-3949.
- FLIR Lepton: <https://www.flir.com.au/products/lepton/>

2.1.3 Project A3: LLM-based Micro-climate Control and Analytics for Indoor Vertical Farms

Maintaining uniform temperature, humidity, and light at canopy level is critical to maximising yield in vertical farming, yet most growers rely on a few wall-mounted sensors that miss shelf-level variations. You will design a distributed BLE sensing network where each node captures temperature, humidity, and PAR every minute. A gateway aggregates 15-minute summaries which are analysed by an LLM (cloud or local) that issues JSON-formatted control commands (e.g. `{"fanB": 90, "ledRow3": -15}`) to fans/LEDs. Demonstrate the closed loop by inducing a hot spot and showing the system corrects it within 5 minutes.

References:

- FAST: A Fog Computing Framework for Industrial IoT—Temperature and Humidity Sensing, IEEE TII, 2019.
- <https://verdantautonomics.com/>

2.1.4 Project A4: Leaf Moisture Monitoring with mmWave Radar

Accurate plant-health monitoring requires measuring leaf water content (WC), but traditional methods clamp sensors onto individual leaves, which prevents large-scale deployment. Recent work shows that mmWave radar can sense leaf WC remotely by analysing reflected signals. You will build a system that combines a TI mmWave radar with environmental sensors (temperature, humidity, light) and a camera to track plant growth and water content non-invasively, validated against a contact-based leaf psychrometer or gravimetric measurement.

References:

- Cardamis, Mark, et al. "Leafeon: towards accurate sensing of leaf water content for protected cropping, with mmWave radar." *IEEE Internet of Things Journal* (2025).

2.1.5 Project A5: RGB LED Bulbs for Communication, Harvesting and Sensing in Smart Farming

RGB LED bulbs are emerging as alternatives to phosphor-coated LEDs for general illumination. In this project you will explore RGB LED lighting for IoT smart-farming applications, performing three roles: (i) visible-light communication (VLC) between nodes; (ii) energy harvesting via a co-located solar cell; (iii) plant-health sensing using reflected light spectra. The deliverable is a single integrated bench-top demo where one RGB bulb illuminates a plant while transmitting data and harvesting energy.

References:

- Pathak, Parth H., et al. "Visible light communication, networking, and sensing: A survey, potential and challenges." *IEEE communications surveys & tutorials* 17.4 (2015): 2047-2077.

2.1.6 Project A6: SweetFruit – Extending Multimodal Fruit Sugar Estimation to Pears

Accurate estimation of fruit sugar content is important for quality control, grading, and supply-chain optimisation in modern agriculture. Recent work such as SweetFruit demonstrated that low-cost multimodal sensing using Near-Infrared (NIR) spectroscopy and Time-of-Flight (ToF) depth sensing can estimate fruit sweetness (Brix value) in real time using machine learning models. In this project, you will extend the SweetFruit system to support pears in addition to apples and strawberries. The will involve collecting a new multimodal dataset of pears using the existing SweetFruit IoT platform. You will then evaluate whether the existing deep learning pipeline can successfully classify and predict pear sugar content.

References:

- SweetFruit: A Two-Stage Mobile Sensing System for Real-Time Fruit Sugar Estimation. The paper can be downloaded on WebCMS3.

2.1.7 Project A7: Mini Growth Chamber for Seed Germination Monitoring

Seed germination is highly sensitive to environmental conditions such as temperature, humidity, airflow, and light exposure. Commercial growth chambers are often expensive, motivating the need for compact and low-cost IoT alternatives. In this project, you will design and build a fully controlled mini growth chamber approximately the size of a microwave oven for monitoring and optimising seed germination. The system will integrate environmental sensors and actuators to maintain ideal growing conditions while providing remote cloud-based monitoring using Home Assistant. The project may include integrating sensors for temperature, humidity, light, and soil moisture, along with environmental control systems such as

heaters, fans, humidifiers, and LED grow lights. A camera module will periodically capture images for cloud-based monitoring and time-lapse analysis. Sensor data and images will be uploaded via MQTT to Home Assistant dashboards for real-time visualization and analytics. You will evaluate the system based on environmental control accuracy, reliability, energy consumption, and germination performance under different conditions.

References:

- Mota-Pérez, Oscar, et al. "Development of a Smart Germination Chamber for Lettuce Seedlings: Integration of Environmental Monitoring and Automated Fertigation Control." International Workshop IFToMM for Sustainable Development Goals. Cham: Springer Nature Switzerland, 2025.

2.2 Category B: Healthcare & Wellbeing

2.2.1 Project B1: Hand Washing Quality Monitoring with mmWave Radar

Hand washing is one of the best ways to stop viruses spreading—if done correctly. Yet only an estimated minority of people follow the WHO 6-step technique. In this project you will build a privacy-preserving device-free system using a TI mmWave radar that detects (i) how many times per day a person washes their hands and (ii) the quality of each event. Develop an associated mobile app that reports statistics in real time and coaches the user with step-by-step corrections.

References:

- Khamis, Abdelwahed, et al. "RFWash: A weakly supervised tracking of hand hygiene technique." Proceedings of the 18th ACM Conference on Embedded Networked Sensor Systems. 2020.
- Nirmal, Isura, et al. "Improving mmwave based hand hygiene monitoring through beam steering and combining techniques." Proceedings of the 23rd ACM Conference on Embedded Networked Sensor Systems. 2025.

2.2.2 Project B2: Assistive Navigation for Vision-Impaired People (Smart Cane / Belt / Glasses)

Vision impairment affects over 357,000 people in Australia and more than 285 million globally, and independent mobility remains one of their biggest daily challenges. The traditional white cane gives only short-range tactile feedback and cannot identify what an obstacle is, warn of head-height hazards, or provide turn-by-turn wayfinding. In this project you will design and build a wearable assistive-navigation aid in one of three form factors via 3D printing – an instrumented *cane*, a *belt*, or *glasses*; and evaluate how well it helps a

(blindfolded) user move safely. A typical system fuses ranging sensors (ultrasonic, depth, or mmWave) with an RGB camera for obstacle detection and a phone for outdoor wayfinding and conveys information through *non-visual feedback*: directional vibration motors and/or concise spoken cues. You will need to validate the system via building a controlled obstacle course in a CSE corridor with known hazards (boxes, hanging objects, a “kerb”); run blindfolded team members; measure collision count and course-completion time from the intended path against a white-cane baseline; collect qualitative feedback.

References:

- Wang, Hsueh-Cheng, et al. "Enabling independent navigation for visually impaired people through a wearable vision-based feedback system." 2017 IEEE international conference on robotics and automation (ICRA). IEEE, 2017.

2.2.3 Project B3: Toothbrushing Quality Monitor using a Wristwatch

Daily toothbrushing is essential for oral health, but home users have very little feedback on whether they are doing it well. In this project you will build a system that monitors brushing on all 16 tooth surfaces using a manual toothbrush and an off-the-shelf smartwatch (or Arduino Nano 33 BLE Sense + magnetometer). Small magnets attached to the brush handle let the wristwatch’s magnetic sensor capture orientation; the IMU and microphone provide complementary motion and acoustic cues for surface recognition.

References:

- Huang, Hua, and Shan Lin. "Toothbrushing monitoring using wrist watch." Proceedings of the 14th ACM Conference on Embedded Network Sensor Systems CD-ROM. 2016.

2.3 Category C: Security, Privacy & Smart Spaces

2.3.1 Project C1: Gait-Based Authentication on Wearable Devices

A person can be identified from their walking pattern recorded by an IMU. In this project you will develop a TinyML biometric authentication method that runs on the Arduino Nano 33 BLE Sense. The system observes the wearer’s gait and decides whether they match the enrolled user. Unlike Project B3, gait authentication does not work when the user is stationary, so consider how the system enrolls, re-authenticates over time, and handles spoofing.

References:

- Derawi, Mohammad Omar, et al. "Unobtrusive user-authentication on mobile phones using biometric gait recognition." 2010 Sixth International Conference on Intelligent Information Hiding and Multimedia Signal Processing. IEEE, 2010.

- Xu, Weitao, et al. "Gait-watch: A context-aware authentication system for smart watch based on gait recognition." Proceedings of the Second International Conference on Internet-of-Things Design and Implementation. 2017.

2.3.2 Project C2: Heartbeat-Based Authentication on Wearable Devices

A gait-based authentication system fails when the person is static (e.g., sleeping). In this project you will use the IMU and pressure sensor in the Arduino Nano 33 BLE Sense to capture ballistocardiogram (BCG) and PPG-like signals, extract heart-rate variability features, and authenticate a seated/lying user.

References:

- Wang, Lei, et al. "Unlock with your heart: Heartbeat-based authentication on commercial mobile phones." Proceedings of the ACM on interactive, mobile, wearable and ubiquitous technologies 2.3 (2018): 1-22.

2.3.3 Project C3: Privacy-Preserving Building Occupancy Monitoring

CSE building managers want occupancy numbers for teaching spaces and need to keep occupant density compliant. Traditional Wi-Fi/BLE sniffing has been undermined by MAC randomisation in modern smartphones. In this project you will build a privacy-preserving occupancy monitor that fuses CO₂ (e.g. SCD41), PIR motion, and optionally mmWave radar—no MAC tracking required. Aggregating data from multiple devices, your system should estimate occupancy levels in a given area, and you may explore how multiple nodes can localise individuals without identifying them.

References:

- Schauer, Lorenz, Martin Werner, and Philipp Marcus. "Estimating crowd densities and pedestrian flows using wi-fi and bluetooth." MobiQuitous. 2014.

2.3.4 Project C4: Real-Time Body Mesh with mmWave Radar

You will build a real-time body-motion monitoring system using a single mmWave radar (no cameras, hence privacy-preserving). The system feeds the radar's 3D point cloud through a deep model that produces a parametric body mesh (e.g. SMPL), which is rendered as an avatar in a mobile app. Applications include sports coaching and home activity monitoring without RGB video.

References:

- Xue, Hongfei, et al. "mmMesh: Towards 3D real-time dynamic human mesh construction using millimeter-wave." Proceedings of the 19th annual international conference on mobile systems, applications, and services. 2021. Codes are available at: <https://github.com/HavocFixer/mmMesh>.

2.4 Category D: Emerging Topics & Robotics-Related Projects

2.4.1 Project D1: Language-Guided Plant Care with a Robotic Arm

Caring for plants in an indoor farm involves many repetitive manipulation tasks, such as moving pots under grow lights, nudging leaves aside, or pointing a nozzle at a target plant. Traditionally each task requires hand-coded perception and motion routines. Vision-Language-Action (VLA) models offer an alternative, mapping a camera image plus a natural-language instruction directly to robot actions. In this project you will deploy an open-source VLA model on an inexpensive desktop arm (SO-101, LeRobot ecosystem) to perform simple language-guided plant-care tasks, e.g. "move the seedling tray closer to the light" or "point at the plant that looks unhealthy." You will collect a few teleoperation demonstrations, lightly fine-tune the model, and evaluate action success rate, end-to-end latency, and robustness to variations such as lighting (e.g. red/blue grow light) or pot placement.

References:

- Shukor, Mustafa, et al. "SmolVLA: A Vision-Language-Action Model for Affordable and Efficient Robotics." arXiv preprint arXiv:2506.01844 (2025).
- Kim, Ju-Young, et al. "Explainable Adversarial-Robust Vision-Language-Action Model for Robotic Manipulation." MobieSec 2025 (poster). arXiv preprint arXiv:2512.11865 (2025).

2.4.2 Project D2: Non-Intrusive Milk Spoilage Detection via Smartphone Vibration

Around one sixth of dairy products are wasted globally each year, partly because printed "best before" dates are rough estimates that ignore real shipping and storage conditions – milk is often discarded while still safe or occasionally consumed after it has spoiled. Laboratory pH meters are accurate but bulky and intrusive and require opening the carton. As milk spoils, bacterial fermentation changes its physical properties (density, viscosity, surface tension), which alters how the liquid responds to vibration. In this project you will build a system that uses the built-in vibration motor and IMU of a commodity smartphone (taped to a sealed carton) to excite and measure the carton's vibration response and a time-series deep-learning model to classify the milk's spoilage stage / pH level. You will need to leave several cartons of milk to spoil at room temperature, use a low-cost pH probe as ground truth, and collect vibration data each time the pH drops by a fixed step; evaluate it as a multi-class pH classifier and, most importantly, as a binary fresh-vs-spoiled detector. Consider robustness across

different phone models (x-axis vs z-axis vibration motors), carton materials, and fill levels.

References:

- Wu, Yuezhong, et al. "VibMilk: Nonintrusive Milk Spoilage Detection via Smartphone Vibration." *IEEE Internet of Things Journal* 11.10 (2024): 17184-17197.
- Huang, Yongzhi, et al. "Vi-liquid: unknown liquid identification with your smart-phone vibration." *Proceedings of the 27th Annual International Conference on Mobile Computing and Networking*. 2021.

2.4.3 Project D3: LLM-Driven Automated Embedded System Development

Throughout the COMP6733 labs you have used *MicroPython* on the Arduino Nano 33 BLE Sense—a high-productivity environment, but one that gives up some of what C/C++ offers: faster execution, smaller memory footprint, lower power per task, finer control over peripherals, and access to the much larger Arduino C/C++ library ecosystem. The traditional barrier to C/C++ is the steeper learning curve: developers must resolve library dependencies, write code that correctly composes sensors with the right APIs, and iterate through compile-and-flash cycles to debug. Recent work shows that Large Language Models can automate this entire pipeline: a user describes a task in natural language, and the system selects libraries, generates the C/C++ sketch, compiles, flashes, reads back debug output, and self-corrects.

In this project you will build a scaled-down LLM-driven automation tool that lets a user write C/C++ for the Arduino Nano 33 BLE Sense (and optionally one other lab board) *from natural language*, using the on-board sensors (LSM9DS1 IMU, MP34DT05 microphone, HTS221 humidity, LPS22HB pressure, APDS-9960 gesture/colour). You will design (i) a prompt that injects per-module API knowledge into the LLM (the original paper does this via a structured “API table” parsed from each library’s header and example files), and (ii) an outer feedback loop that captures `arduino-cli` compile errors and serial debug output, and feeds them back to the LLM for the next attempt. You will need to build a small “task bench” of 30–50 natural-language requests at three difficulty levels, for example, single-module (read a sensor and print), two-module integration (sensor triggers an actuator), and three-module composition (sensor + actuator + BLE notification); and report coding accuracy (does the generated sketch compile and use the correct APIs) and completion rate (does the flashed system actually do what the user asked, verified via debug output or a physical observation). Compare your full pipeline against a no-feedback-loop baseline (direct LLM prompting, no compile/flash iteration) to isolate how much the feedback loop actually helps. Also compare at least two LLMs (e.g. a smaller/cheaper model vs. a frontier one) and report latency and token cost.

References:

- Yang, Huanqi, et al. "AutoEmbed: LLM-driven Automated Software Development for Generic Embedded IoT Systems." Proceedings of the 2026 ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems. 2026.

2.5 Project X: Proposed your own IoT project

3 General Requirements

3.1 Choice of project

You can do one of the proposed projects or propose your own project. If you want to propose own project, please make an appointment to speak with the LiC with your proposal in Week 2 or 3.

3.2 Formation of project teams and bidding process

You are expected to do this project in a group of students. You are free to form your own team. Please enter your team membership using the WebCMS function available at the course website: <https://webcms3.cse.unsw.edu.au/COMP6733/26T2/groups/>.

Please email cs6733@cse.unsw.edu.au your team membership or list it in WebCMS (please make your group visible) by **17:00, Wednesday 17 June 2026**. In your email, please also include your preferences for each of the proposed projects. You are asked to express your preference by giving each project a score of 1, 2 and 3, **where ‘1’ means a project that you most want to do and ‘3’ means a project that you least want to do**. Please express your preference for all the projects. You can have any combination of 1 to 3, e.g., you can give all projects a ‘3’ or you can give 2 projects a ‘3’ and a project a ‘1’. The LiC will attempt to allocate the projects according to your preference. The aim is to attempt to balance the number of groups doing each project. If we do not receive your team membership by this time, we will do a random team formation for you.

The LiC will attempt to allocate the projects according to your preference. The aim is to attempt to balance the number of groups doing each project.

If we do not receive your team membership by this time, we will do a random team formation for you.

3.3 Assessment

Note the following assessment and submission requirements (one report from each group please):

1. You should submit a preliminary report (with your proposed solution, project plans, and milestones for both Week 8 and the final demonstration) by **17:00 Friday 3 July 2026** via the give system. The report will be assessed based on the quality of the research that you have done, your explanation on why your proposed solution best fit the problem and the quality of your work plan. The give command to use is: “give cs6733 projectPlan”. Your project report must be in PDF format. Please tar your report and call it “project0.tar”. This is the only filename accepted by the system.
2. In Week 5, each team will give a presentation on their proposed solution and their proposed work plan to the whole class. Each team member should do a part of the presentation. LiC and fellow students will assess the presentations according to predefined marking criteria.
3. In Week 8, each team will give a presentation on their project progress and demonstrate its project milestones to the whole class. The aim is to ensure that you have made progress on the project. Each team member should do a part of the presentation. LiC and fellow students will assess the presentations according to predefined marking criteria. In Week 10, each team will need to negotiate with the LiC on a time to demonstrate. The milestone report is to be submitted by **17:00 Friday 24 July 2026** via give. The give command to use is: “give cs6733 projectMilestone”. Please tar your final submission and call it “project1.tar”. This is the only filename accepted by the system.
4. The project report and code are to be submitted by **17:00 Friday 10 August 2026** via give. The give command to use is: “give cs6733 projectFinalReport”. Your project report must be in PDF format. Please tar your final submission and call it “project2.tar”. This is the only filename accepted by the system.
5. A final demonstration and interview will be held on **11 August 2026**. Each team is given approximately 30 minutes for the demo, and this will also include a short interview with each student. You will be asked to express preferences on the interview day later on.
6. **Peer Assessment** Each student will be required to evaluate their team member’s contributions to the project, including system design, implementation, testing, documentation, and related project management tasks, via Moodle. If contributions were uneven, please include a brief explanation in the comments. The Moodle module will be available from 31 July 2026 until 12 August 2026.

The project in total is worth 60% of your final mark in this course. The breakdown is as follows:

1. Project Plan and preliminary report + Class Presentation in Week 5: 10%

2. Project progress demo + class presentation in Week 8: 10%
3. Project final report: 15%
4. Project final demo: 25%

The deadlines are hard. No extensions will be offered.